

GAIRA

Pure Ag-SERS Abstraction Recovery (V5)

From exact molecular identity to recoverable biochemical abstraction. When exact identity is lost after Ag-SERS, does the correct broader chemistry — component, motif, subclass, theme — still survive?

Frozen atlas fingerprint 09ed804a40836f4a05a91ba10900cded · reproduces V4 exact identity (7/3/4)

Subclass is an EVALUATION OVERLAY, never a new GAIRA axis · SERS validates, never trains

Headline: expected motif/theme often PRESENT (top-3: MSS 40%, theme 49%) but rarely SPECIFIC (2/48, 1/51); cross-modal class recovery at chance; only functional perturbation (3) recovers class chemistry beyond exact identity.

1 · Executive summary

When exact analyte identity is lost after Ag-SERS, does the correct broader chemistry survive? The expected motif/theme is often PRESENT in the Ag-SERS top-3 (MSS 19/48 = 40%, theme 25/51 = 49%), but that presence is **not analyte-specific**.

Specific, null-and-background-adjusted recovery stays rare at every level (component 2/51, MSS 2/48, theme 1/51), and cross-modal subclass/family classification is AT CHANCE (balanced accuracy 0.03-0.18, all permutation p n.s.). **Presence ≠ recovery**.

A Raman→Raman control proves the taxonomy is separable and that abstraction DOES help within Raman (subclass 0.23 → family 0.35 → theme 0.42) — the Ag-SERS modality gap collapses it. The failure is the modality, not the taxonomy or the method.

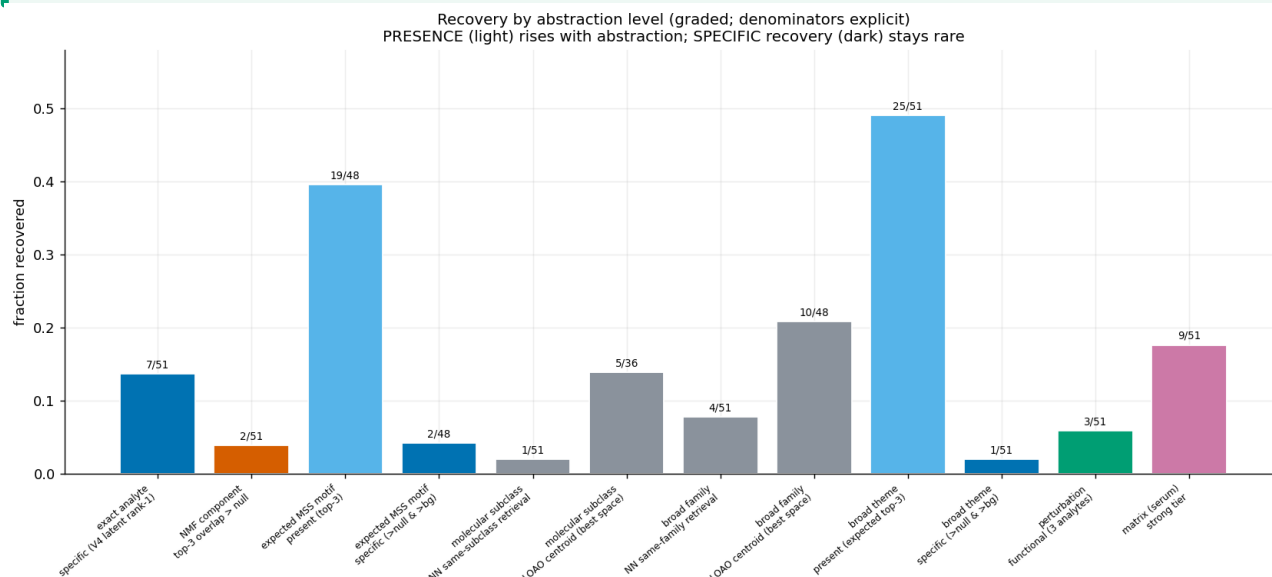


Figure. Graded recovery: PRESENCE (light) rises with abstraction; SPECIFIC recovery (dark) and classification (grey, at chance) stay low.

2 · The decisive control — abstraction helps in Raman, collapses across the modality gap

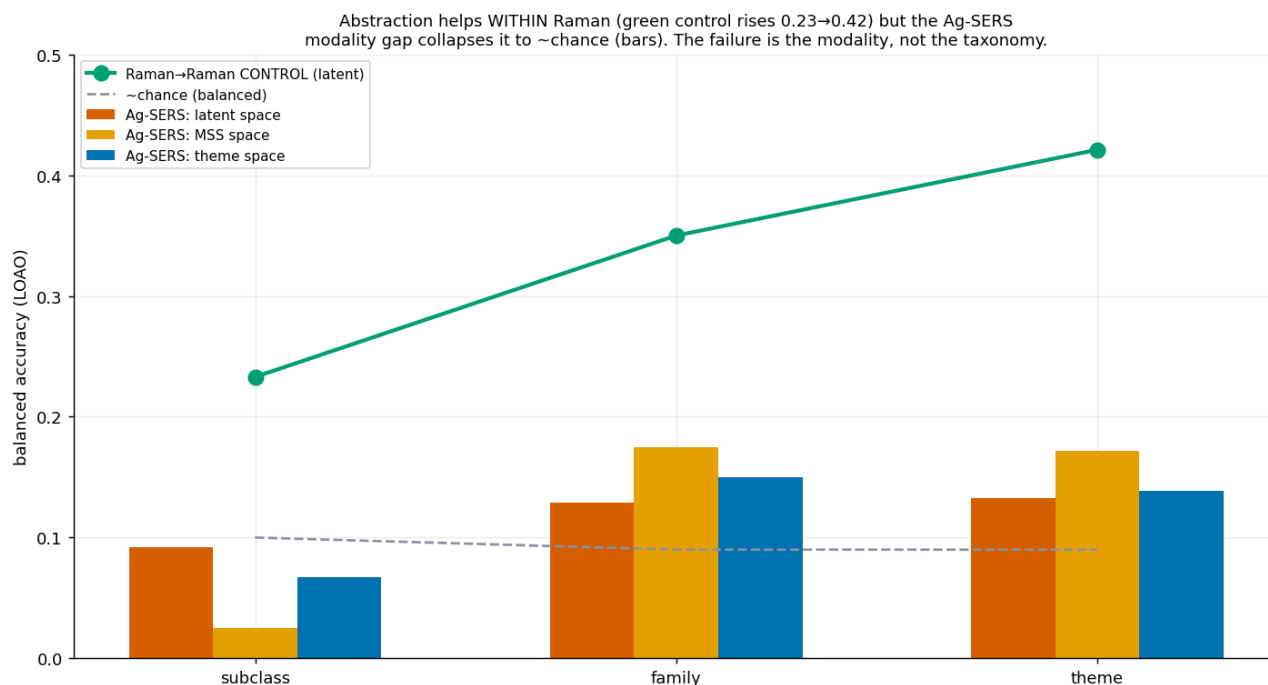


Figure. Raman→Raman control (green) rises with abstraction and sits well above chance; Ag-SERS cross-modal (bars) sits near chance.

This isolates the cause: the chemical taxonomy IS separable and abstraction genuinely improves recovery — but only within the Raman training modality. The Ag-SERS surface reshaping (the purine attractor and adsorption selection) destroys the class structure, so cross-modal subclass/family recovery is at chance.

3 · Recovery ladder & classification

Level	Tier	n	denom	frac
exact analyte	specific (V4 latent rank-1)	7	51	0.137
NMF component	top-3 overlap > null	2	51	0.039
expected MSS motif	present (top-3)	19	48	0.396
expected MSS motif	specific (>null & >bg)	2	48	0.042
molecular subclass	NN same-subclass retrieval	1	51	0.020
molecular subclass	LOAO centroid (best space)	5	36	0.139
broad family	NN same-family retrieval	4	51	0.078
broad family	LOAO centroid (best space)	10	48	0.208
broad theme	present (expected top-3)	25	51	0.490
broad theme	specific (>null & >bg)	1	51	0.020
perturbation	functional (3 analytes)	3	51	0.059
matrix (serum)	strong tier	9	51	0.176

gran	space	bal acc	macro-F1	acc
subclass	latent	0.092	0.111	0.111
subclass	MSS	0.025	0.040	0.028
subclass	theme	0.067	0.087	0.083
family	latent	0.129	0.118	0.125
family	MSS	0.175	0.115	0.146
family	theme	0.150	0.092	0.125
theme	latent	0.133	0.090	0.184
theme	MSS	0.172	0.126	0.184
theme	theme	0.139	0.086	0.122
subclass	latent(Raman→Raman control)	0.233	0.239	0.278
family	latent(Raman→Raman control)	0.350	0.371	0.417
theme	latent(Raman→Raman control)	0.421	0.418	0.510

4 · Per-analyte ladder & highest recovered level

Per-analyte recovery ladder — dark ✓ = specific, light ✓ = present-only, · not tested, u unassigned
sorted by family; presence-only cells dominate

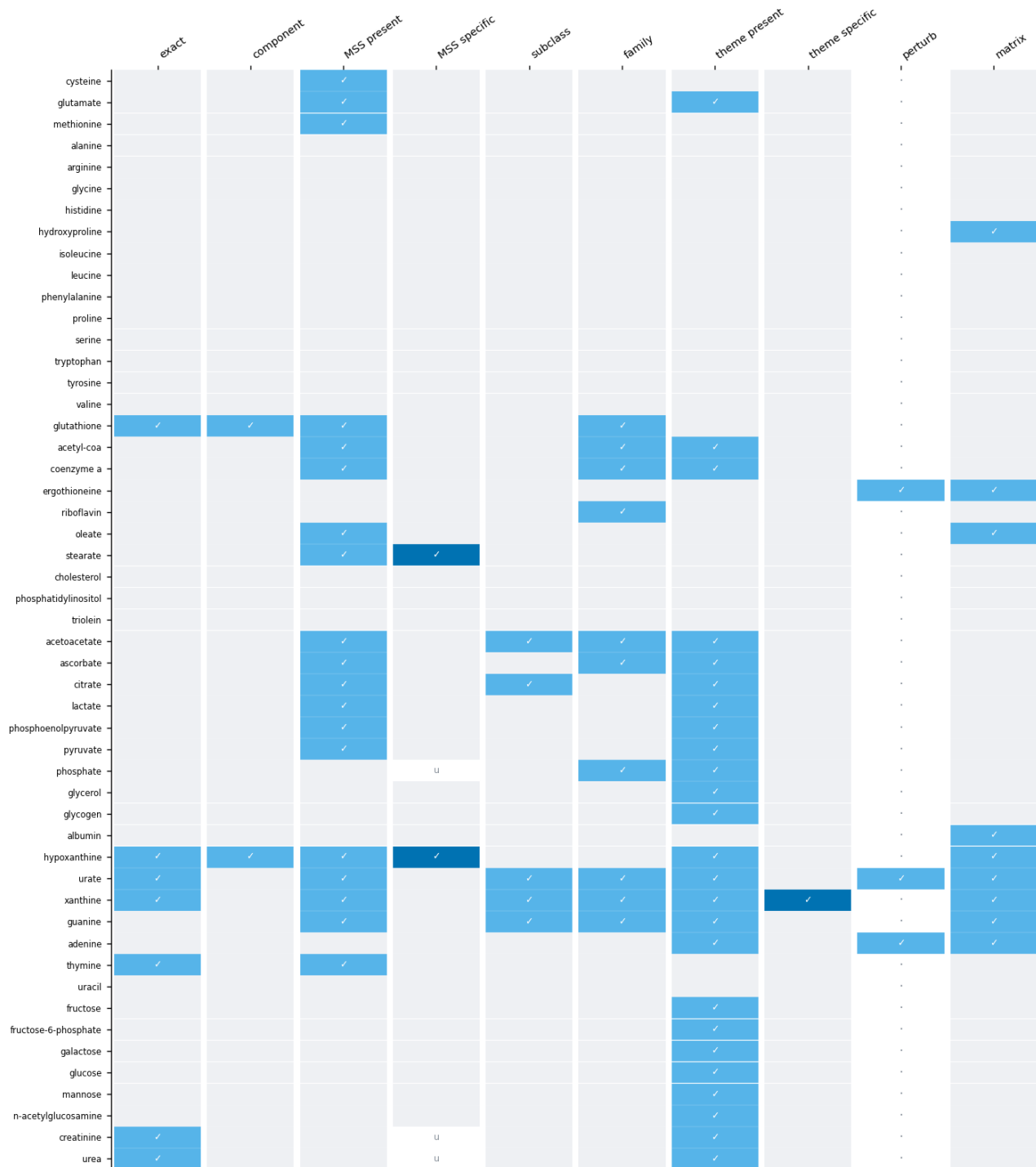


Figure. Per-analyte ladder — dark ✓ specific, light ✓ present-only, · not tested, u unassigned. Presence-only cells dominate.

5 · MSS motif & broad theme recovery

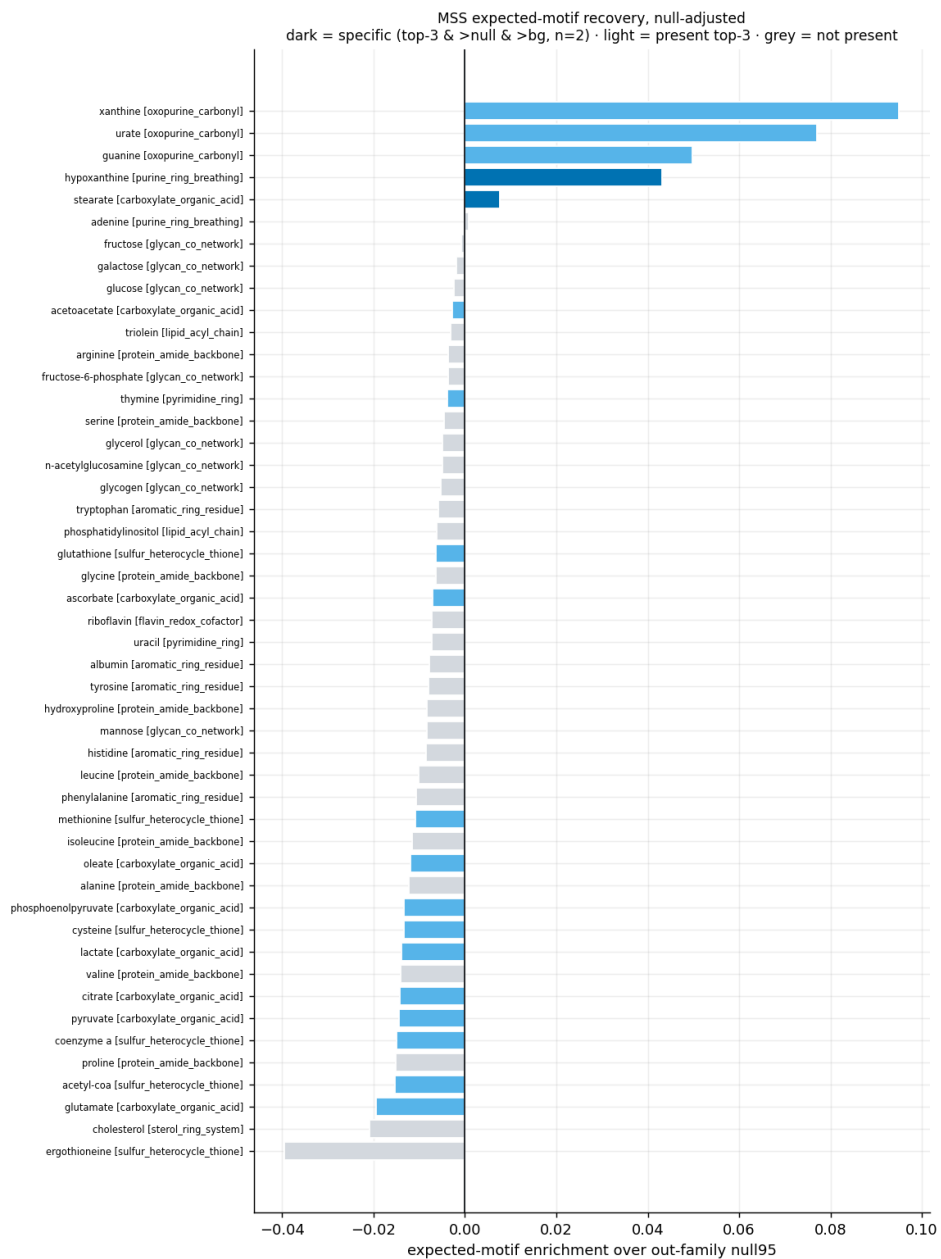


Figure. MSS expected-motif recovery, null-adjusted (dark = specific, n=2; light = present).

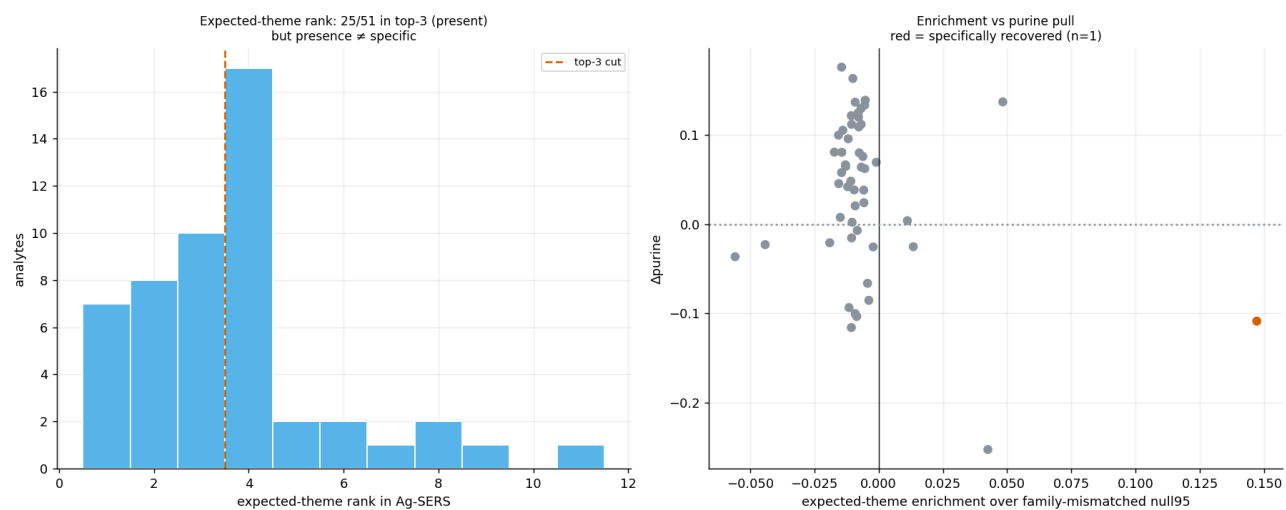


Figure. Expected-theme rank (25/51 top-3) and enrichment (1 specific).

6 · Purine attractor, perturbation & matrix

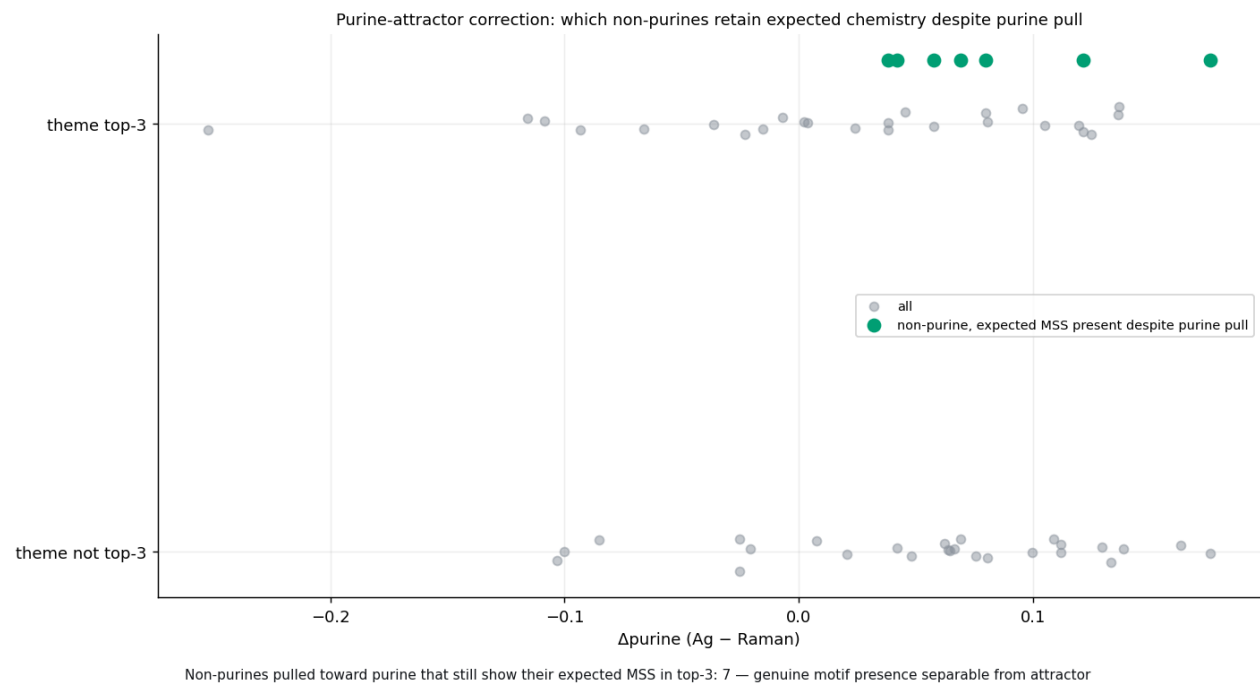


Figure. Non-purines that retain expected chemistry despite purine pull — genuine presence vs attraction.

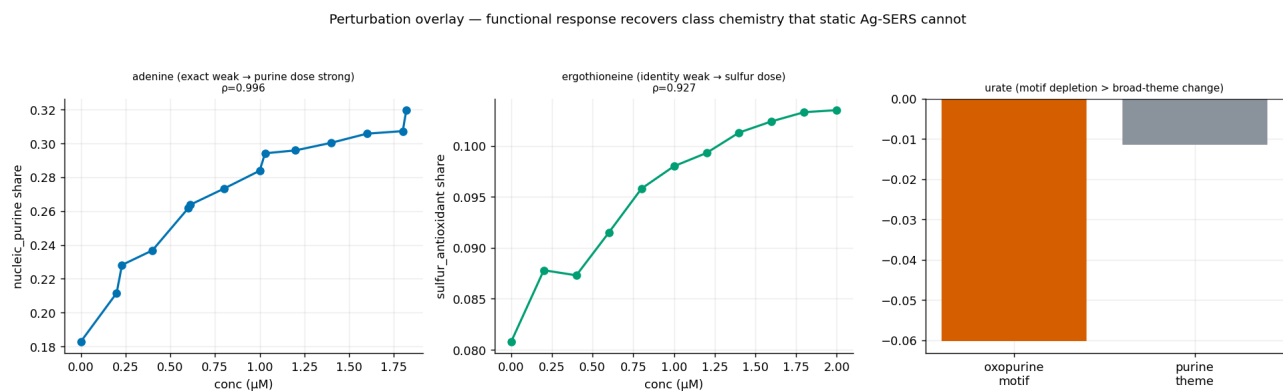


Figure. Functional perturbation recovers class chemistry static Ag-SERS cannot (adenine, ergothioneine, urate).

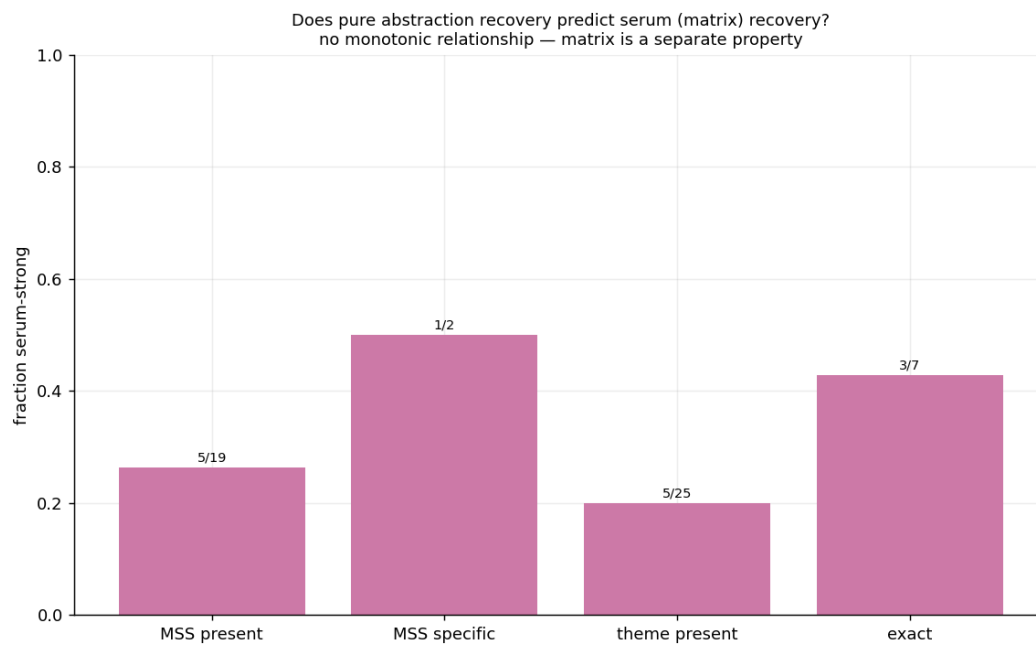


Figure. Pure abstraction recovery does not predict serum strength — matrix is a separate property.

7 · Representative analytes

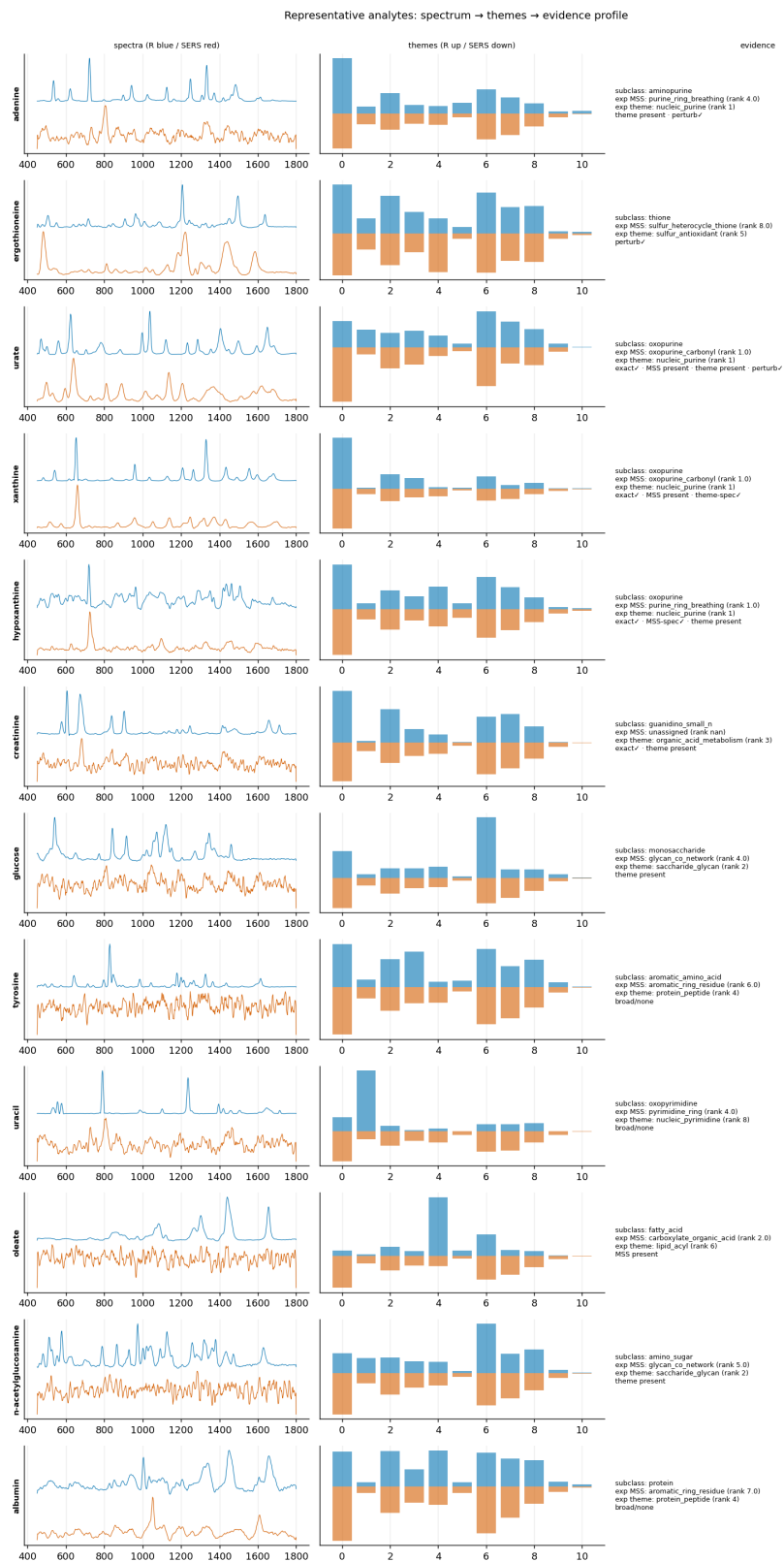


Figure. Spectrum → themes → evidence for 12 representatives.

8 · What GAIRA can & cannot claim · limitations · conclusions

Can: report a broad theme/motif is PRESENT (top-3) for ~40–50% (broad interpretation); specifically recover exact identity for a strong-chemisorber minority (7/51); functionally validate 3 analytes. **Cannot:** claim molecular identification from motif/theme presence; claim class-discriminative subclass/family recovery from pure Ag-SERS (at chance); infer perturbation/matrix for untested analytes.

Limitations. Raman-trained atlas; surface-selection effects; purine attractor; low exact identity; subclass imbalance (15 exploratory singletons); 3 perturbation cases; no Au-SERS observation model; confidence $\neq$ identifiability; cross-modal centroid classification carries the global-shift confound (reported with the Raman control).

Conclusions. Broad biochemical presence frequently survives Ag-SERS, but it is not analyte-specific or class-discriminative — abstraction raises apparent presence via a shared attractor, while the modality gap collapses genuine class recovery to chance. Specific recovery beyond a strong-chemisorber minority comes only from functional perturbation. **Recommended next experiment:** a learned Raman→SERS observation model, and dynamic perturbation (DART) as the route to class-specific recovery.

Provenance. Frozen atlas 09ed804a40836f4a05a91ba10900cded unchanged; reproduces V4 identity (7/3/4); deterministic; method in EVALUATION_HIERARCHY_AND_METRICS.md; overlay provenance in ANALYTE_CLASSIFICATION_PROVENANCE.md.